PCB AXIAL-FLUX MOTOR

Integrated Electromagnetic and Thermal Modelling: Detailed Explanation and Project Implications

EEE4022S — Final-Year Engineering Project

Technical companion to the 50-page electromagnetic modelling record and the 14-page Icepak thermal analysis report, both dated 8 October 2026.

Purpose. Explain what the modelling established, how the methods work, how the electromagnetic and thermal results relate, what can reasonably be claimed, and what remains uncertain before prototype validation.

Evidence convention. “EM record” refers to PCB Axial-Flux Motor: Modelling Timeline and Methodology Record. “Thermal record” refers to PCB Axial-Flux Motor: Thermal Analysis Report. Derived arithmetic in this document is explicitly labelled and is not represented as a new solver result.

Prepared 8 October 2026. This is an explanatory study document, not a substitute for the raw Maxwell/Icepak runs, solver logs, scripts or formal thesis citations.

Executive synthesis

The electromagnetic (EM) work establishes a validated, computationally feasible method for comparing a coreless PCB axial-flux motor and two cascaded extensions. It predicts fundamental torque constants of 19.496, 41.636 and 60.284 mN·m/A for Rev0, Rev1-S and Rev1-D. The thermal work uses steady-state ANSYS Icepak to assess heating from winding copper losses, imposing a 100 °C board criterion and an 80 °C magnet criterion under a stationary-rotor, assumed-housing test condition. Its originally reported continuous torque capacities are 47, 77 and 111 mN·m, respectively. [EM record pp. 1–2, 34–37; thermal record pp. 1, 8–9]

The studies answer different but complementary questions. EM characterisation says how much torque and back-EMF a current produces. Thermal analysis says what current can be sustained without exceeding a specified temperature under assumed cooling. Combining them is how the project progresses from peak magnetic capability to potential continuous performance.

Reconciliation needed: the thermal report converts current to torque using earlier EM constants (18.02, 38.48 and 56.68 mN·m/A). The final EM report superseded these constants. Consequently the temperature/current solutions and the published torque labels are not based on an identical EM revision. Section 9 below provides transparent arithmetic at the reported current limits; these derived quantities are not new Icepak solves. [Thermal record pp. 9, 14; EM record pp. 1–2, 17, 43]

1. The engineering problem and three configurations

PCB windings eliminate the conventional iron-toothed stator, enabling a thin axial-flux machine, but limit copper cross-section and heat removal. A single rotor field also weakens through a five-board stack. The engineering question is therefore not just whether torque can be generated; it is whether adding a second active stage improves torque production and thermally permissible output enough to justify extra magnets, copper, rotor material and mechanical complexity.

Design	Physical topology	Boards and magnets	Interpretation
Rev0	S1–R1: one five-board stator, one single-sided rotor	5 PCBs; 16 N35 magnets	Baseline: single active rotor interface
Rev1-S	S1–R1–S2–R2: two stators, each rotor single-sided	10 PCBs; 32 magnets	Cascaded comparator; S2 sees steel at one side
Rev1-D	S1–R1(double-sided)–S2–R2; middle rotor carries magnets on both sides	10 PCBs; 48 magnets	Primary cascaded design; magnetic excitation on both sides of S2

Shared geometric inputs include a 12-sided board approximately 93 mm across flats, 25 mm central opening, six 35 µm copper layers, five boards per stator, 1.02032 mm axial pitch, 19 mm shaft, sixteen 20 × 6 × 2 mm N35 magnets per magnetised rotor face, 2 mm steel backing and 1.0 mm nominal active gap. The 16 poles correspond to eight pole pairs. [EM record pp. 3, 37–40; thermal record pp. 2–3]

2. What the electromagnetic modelling actually did

2.1 Input geometry and conductor network

The PCB source was taken through ODB++ and reconstructed in AEDT. Validation addressed conductive-island assignments, vias, pads, phase separation, and external terminals. The record distinguishes 615 unique vias from 617 initial via records after near-coincident duplicate records were identified. The stator build connected board k layer-L6 output to board k+1 layer-L1 input with copper bridges; 36 junction probes in each five-board stator were checked. [EM record pp. 6–9]

A major lesson is that geometry that looks visually continuous does not guarantee valid conduction in a finite-element solver. Internal current terminals initially produced conduction-path errors; the terminal approach was revised to place external leads on the solution-region boundary. Terminal definitions and current-variable creation order then required further checks. These practical corrections became repeatable script-level validation gates.

2.2 Single-board electrical characterisation

Quantity	Reported result	Meaning
Phase resistance A / B / C	0.27776 / 0.27736 / 0.27789 ohm	Balanced DC conduction paths; mean ≈0.2777 ohm per board-phase
Single-board self-inductance	≈2.07 microhenry	Relevant to voltage-driven current response
Two-board phase-series test	0.55313 ohm	Agrees with approximately twice single-board resistance
Five-board phase resistance	1.389 ohm	Resistance used for Rev0 system model; Rev1 doubles to 2.777 ohm

These results validate the electrical representation, not a fabricated PCB measurement. In the later field-only model, an ODB++ nodal track-current network yielded approximately 0.2656 ohm per board-phase, whereas the final system calculation retained the detailed copper-FEM value of 0.2777 ohm. [EM record pp. 7–8, 13, 30, 35]

2.3 Rotor-angle solution and reciprocity

Rather than solve a continuously rotating and copper-meshed transient machine, the detailed reference solved magnetostatic fields at fixed rotor angles with positive and negative applied currents. A reciprocity relationship links torque to the angular derivative of permanent-magnet flux linkage. The extracted position-dependent coefficient also defines back-EMF at angular speed. The record used the copper-side torque for single-current runs because of bias in rotor-side virtual torque. [EM record pp. 8, 29–30]

$$k(\theta) = d\lambda/d\theta = [T(+I) - T(-I)] / (2I)$$

$$e(\theta) = \omega \times k(\theta) \; ; \; T = \sum_{\text{phases}} [i_{\text{phase}} \times k_{\text{phase}}(\theta)]$$

The initial detailed reference found a 6.0917 mN·m/A fundamental and −30.18° electrical phase under the original 5 mm region padding. Its period-repeat, half-wave antisymmetry, copper-torque linearity and torque-comparison checks supported the modelling of a nearly linear, coreless machine. The subsequent boundary correction changed the detailed reference to 6.2816 mN·m/A, so the older

amplitude is a historical validation reference rather than a final headline. [EM record pp. 8, 16–17, 35–36]

3. Why the modelling method changed

The fully copper-meshed five-board motor did not complete on the approximately 16 GB lab PCs. The copper was about 35 µm thick and the inter-board solder-mask spacing about 20 µm, generating extreme mesh scale separation. Three full-Rev0 solve attempts stalled or froze during meshing or initial adaptive solution. The report attributes the principal resource burden to fine tetrahedral meshes, direct matrix factorisation and adaptive refinement. That is a computational limitation, not a demonstration of motor failure. [EM record pp. 31–33]

3.1 The accepted field-only FEM approach

The selected method solves the magnet/steel/air field in Maxwell, then samples the field along winding centre-lines extracted from the ODB++ PCB layout. The tracks are represented by chord segments (0.4 mm) and via steps; a DC network solution determines current in each element. Python then integrates the Lorentz contribution of those current-carrying paths to torque at each prescribed rotor angle. This preserves the actual winding layout without requiring Maxwell to mesh the copper in the full machine. [EM record pp. 12–13, 34–35]

$$\mathbf{dF = I \times (dl \text{ cross } B)\ ;\ T_z = \text{sum over trace/via segments } [(r \text{ cross } dF)_z]}$$

The complete field-only models used 8 angular positions in an electrical period, four adaptive passes, 0.8 mm air-zone mesh near the stators, 2 mm magnet mesh and 20 mm final outer-region padding. Reported sample sizes were 180,775 field points in Rev0 and 361,550 in Rev1. No explicit copper armature field, copper eddy losses or spatial averaging across track cross-section was included. [EM record pp. 34–35]

3.2 Validation: a converged model can still be wrong

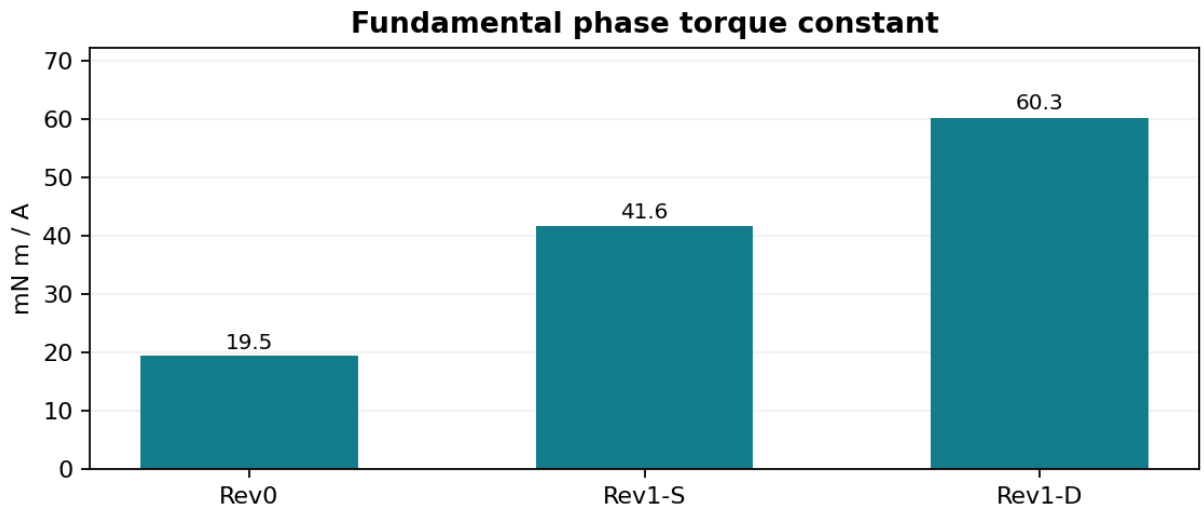
At first, a gate comparing the field-only board with the detailed copper-FEM board failed: the torque-constant difference was about +4.12% when a maximum of 2% was allowed. Refining the field-only zone mesh from 0.8 to 0.55 mm changed the result by just −0.03%; a separate gap-mesh refinement changed a detailed-model result by about −0.01%. The mismatch was traced to the outer solution-region boundary, whose 5 mm padding biased the single-board reference. Using 20 mm padding in both models resulted in 6.2816 versus 6.3697 mN·m/A, a +1.40% difference and 0.04° phase difference, satisfying the 2% / 1° gate. [EM record pp. 14–17, 35–36]

This is a central engineering finding: mesh convergence is necessary but not sufficient; field-boundary position also needs independence testing. The validated Rev0 method should not be described as exact, and the same 1.4% result cannot be automatically asserted for all Rev1 S2 boards.

4. Final electromagnetic comparison

Metric	Rev0	Rev1-S	Rev1-D
Magnets	16	32	48
Fundamental K (mN m/A)	19.496	41.636	60.284
Ideal three-phase torque at 1 A peak (mN m)	29.2	62.5	90.4
Phase back-EMF at 1000 rpm (V rms)	1.444	3.083	4.464
R_phase (ohm)	1.389	2.777	2.777
Motor constant Km (N m / sqrt(W))	0.0203	0.0306	0.0443
Third harmonic / fundamental	5.9%	5.6%	5.8%

Source: EM record pp. 1–2, 17, 36–37, 43. EM values are finite-element-derived model outputs, not prototype measurements.



Rev1-S provides approximately 2.14 times Rev0’s K, and Rev1-D approximately 3.09 times. Unlike an unqualified statement of “better motor”, these ratios refer to electromagnetic coupling and ideal torque-per-ampere at fixed geometry and gaps. Adding stages also adds resistance, magnetic material, inertia and voltage demand. [EM record pp. 2, 43–44]

4.1 Distribution through a five-board stator

PCB position	Rev0 S1 K (mN m/A)	Rev1-D S2 K (mN m/A)
1	6.3697	8.7587
2	4.7962	7.8497
3	3.5881	7.5412
4	2.7268	7.8575
5	2.0154	8.7539
Total	19.496	40.761

The Rev0 stack exhibits diminishing contribution as boards sit progressively farther from its single rotor. By contrast, Rev1-D's S2 lies between magnetised rotor faces and obtains strong contributions near both ends of the stack. This is the physical reason cascading offers more than simply extending one PCB stack. The report also records that S1 and S2 remain effectively in phase, supporting same-sense series wiring. [EM record pp. 36–40]

4.2 Material loading and axial magnetic force

The Rev1-D shared 2 mm steel plate reached a reported peak magnetic flux density of approximately 1.23–1.30 T and was judged unsaturated within the model. Maxwell stress estimated approximately 47–48 N axial attraction between the rotors. This produces a serious requirement for axial stiffness, spacers, bearings and stable 1 mm gaps. The force is an electromagnetic estimate, not a measured mechanical load. [EM record pp. 2, 40]

5. MATLAB: translating EM properties to motor operation

A FEM-informed lumped electromechanical MATLAB model received the final field-only K and harmonic ratios. It considered imposed-current torque/speed response, voltage-driven electrical dynamics and steady-state torque–speed/efficiency envelopes. Its resistance was taken from copper FEM, while inductance, inertia, controller and supply were estimated or assumed. [EM record pp. 41–43]

$$J \times \frac{d\omega}{dt} = T_e - T_{load} - B \times \omega$$

$$L \times \frac{di}{dt} = v_{phase} - v_{neutral} - R \times i - e_{phase}$$

Final MATLAB output	Rev0	Rev1-S	Rev1-D
Mass estimate	227 g	368 g	401 g
12 V no-load speed	3393 rpm	1589 rpm	1097 rpm
12 V speed at 20 mN m	2928 rpm	1385 rpm	1000 rpm
Cu-loss-limited efficiency (20 mN m, 1000 rpm)	68.3%	83.1%	91.1%
0 to 1000 rpm (ideal control, no load)	0.202 s	0.186 s	0.142 s

The benefit of high K is reduced current for a specified torque, but the cost is higher back-EMF at the same speed. At the modelled 12 V bus, Rev1-D has little or no speed margin at the specified loaded 1000 rpm point. The efficiencies above account for copper loss only; they must not be presented as measured total efficiencies. [EM record pp. 43–44]

6. Thermal study: what was solved and why

The thermal analysis treats the stator as the principal practical operating limit because the coreless PCB lacks a massive laminated steel heat sink. The Icepak model solves steady-state solid conduction plus gravity-driven and fan-forced air convection for each design with rotor(s) held still. Loss is applied to an effective winding annulus and updated iteratively with copper temperature. [Thermal record pp. 1–4]

$$P_{Cu} = 3 I_{rms}^2 R_{phase}(T) = (3/2) I_{pk}^2 R_{phase}(T)$$

$$R(T) = R_{20} \times [1 + 0.00393 \times (T - 20\ °C)]$$

Both the electrical and thermal calculations employ R20 = 0.2777 ohm per board-phase, or 1.389 / 2.777 ohm per machine-phase for Rev0 / Rev1. Resistance feedback matters because loss increases as the winding heats, creating a nonlinear rise in temperature at fixed current. [Thermal record p. 2]

6.1 Thermal geometry, materials and assumptions

The housing is a provisional test-bench assumption: a sealed acrylic tube (120 mm inner diameter, 3 mm wall, 60 mm length), aluminium end plates/base, tie rods, shaft and effective bearing heat-transfer rings. The hood air is set to 30 °C, with fans modelled as 1 m/s axial flow in the on cases. Gravity acts downwards. Radiation is excluded from Icepak; the rotor remains stationary. [Thermal record pp. 3–4]

The boards use homogenised anisotropic thermal conductivity (40.3 W/m K in plane, 0.334 W/m K through the thickness), representing six thin copper layers at an assumed 50% coverage. Steel is 52 W/m K, magnets 9 W/m K, acrylic 0.19 W/m K, aluminium 167 W/m K. Ten-board Rev1 designs were meshed with each five-board stator merged into one effective body and winding annulus after the thin-board mesher omitted a winding. Per-board values are recovered using location monitors. [Thermal record pp. 4–5, 10]

6.2 Temperature-resistance iteration

Icepak was called from PyAEDT scripts. At a current setpoint, copper loss was calculated from an assumed temperature, Icepak solved temperature, resistance and loss were updated, and the case was repeated to reach approximate consistency. The record says the target was a loss change below 1%, normally reached after 2–5 solves. Its verification table reports last-step loss changes of 0.2–1.1% across cases; the upper end slightly exceeds the stated target and should be checked before a claim that all cases strictly converged below 1%. [Thermal record pp. 4, 10]

7. Thermal results: Rev0

Case	Cu loss	Hottest winding	Magnet indicator
0.737 A peak, fans on	1.22 W	40.9 °C	36.8 °C
1.0 A peak, fans on	2.32 W	50.3 °C	42.7 °C
2.0 A peak, fans on	11.57 W	124.9 °C	89.2 °C
2.0 A peak, fans off	11.87 W	134.5 °C	98.0 °C

At the original EM-calibrated 20 mN m current of about 0.737 A peak, Rev0's hottest winding is approximately 41 °C in the stationary-rotor, fans-on simulation. At 2 A peak, the stationary rotor's board exceeds the selected 100 °C limit, and the magnets exceed their nominal 80 °C rating. The innermost/hottest stack positions are not identical: the board next to the rotor is generally cooler, since it has an additional thermal transfer path across the air gap; boards deeper in the stack become hotter. [Thermal record p. 5]

The thermal resistance inferred from the winding-to-hood rise is approximately 8.0–8.8 K/W in the simulated Rev0 cases. The assumed external fans provide only modest cooling because much of the limiting thermal resistance remains inside the closed housing. [Thermal record p. 5]

7.1 Dynamic heating and rotor stirring: separate reduced model

A 12-node lumped thermal network was calibrated against still-rotor Rev0 Icepak temperature rises to roughly ±7%, then used to estimate warming and additional airflow from rotation. The model estimates that, at 2 A peak with a locked rotor, the hottest board reaches 60 °C after about 6 min, 80 °C after 15 min and 100 °C after 37 min; rotating at 1000 rpm instead yields about 82 °C steady under the network assumptions. [Thermal record pp. 6–7]

For Rev0, the network's predicted continuous current rises from approximately 1.77 A peak (rotor locked, fans on) to 2.25 A peak (1000 rpm, fans on). This is not an Icepak rotating-wall solution or measured motor rating. The empirical stirring correlation is identified as the largest uncertainty and should be calibrated on the rig. [Thermal record pp. 6–7]

8. Thermal results: Rev1-S and Rev1-D

Stationary-rotor case	Rev1-S	Rev1-D
At 20 mN m (original EM K)	0.345 A; 34.2 °C hot spot; 0.52 W	0.234 A; 32.0 °C hot spot; 0.24 W
1 A peak, fans on	65.7 °C hotspot; 4.84 W	67.1 °C hotspot; 4.87 W
2 A peak, fans on	220.4 °C hotspot; 28.9 W	233.4 °C hotspot; 29.8 W

Rev1-S and Rev1-D run at relatively similar temperatures at the same current, because both contain ten boards and have similar total resistive losses. Rev1-D becomes attractive at the same demanded torque, because its larger K requires less phase current and therefore less I^2R loss. At 2 A peak, both Rev1 designs exceed reasonable board and N35 magnet temperature limits by a very large margin in the stationary-rotor simulation. [Thermal record pp. 7–8]

The model predicts the first stator, S1, runs slightly hotter than S2. The original thermal study reports that the Rev1 stack-to-hood thermal resistance is around 6.6–7.6 K/W, somewhat lower than Rev0 due to greater surface area, but not halved because both stators remain in one enclosure. [Thermal record pp. 8, 10]

9. Electromagnetic–thermal integration and reconciliation

9.1 Two EM revisions appear in the reports

The final EM record explicitly supersedes the earlier board-by-board machine K values. The thermal report, produced in parallel, uses an earlier set to choose currents and label torque axes. This is a version-of-input inconsistency, not evidence that the solver temperature outputs have been recalculated. The three Icepak-derived current limits should therefore be preserved as the results of the reported thermal model; torque at those currents can be recalculated arithmetically for comparison. [EM record pp. 1–2, 17; thermal record pp. 9, 14]

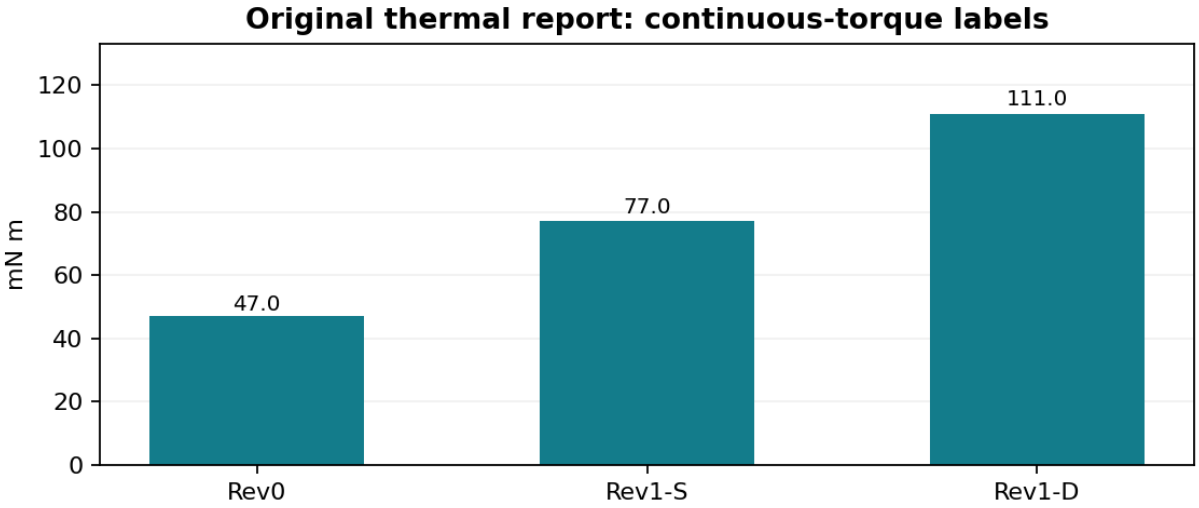
Motor	Original thermal K	Final EM K	Thermal current limit
Rev0	18.02 mN m/A	19.496 mN m/A	1.75 A peak
Rev1-S	38.48 mN m/A	41.636 mN m/A	1.33 A peak
Rev1-D	56.68 mN m/A	60.284 mN m/A	1.31 A peak

9.2 Illustrative torque conversion using the final K

$$T_{mean,\ ideal} = 1.5 \times K_{final} \times I_{peak}$$

Motor	Thermal report torque	Derived with final EM K	Status
Rev0	47 mN m	51.2 mN m	Derived; not a new Icepak result
Rev1-S	77 mN m	83.1 mN m	Derived; not a new Icepak result
Rev1-D	111 mN m	118.5 mN m	Derived; not a new Icepak result

The derived entries hold the thermal report’s stationary-rotor current limits fixed and apply the final EM fundamental torque constants. They are conditional comparisons, not confirmed continuous torque ratings. Thermal resistances, geometry, heat-source distribution, material limits and numerical convergence would still need to be checked at the final selected test operating points. In particular, Rev1-S2 absolute EM predictions retain an unresolved 3–11% difference against separate copper-resolved reference runs. [EM record pp. 36–37; thermal record pp. 9–11]



9.3 Thermal-limit current is not identical to drive current limit

The MATLAB comparison assumes a 2 A peak drive-current ceiling. Icepak predicts lower sustained currents for a stationary rotor, particularly in Rev1-S and Rev1-D (1.33 and 1.31 A peak, respectively). Hence ideal MATLAB peak performance at 2 A must not be presented as indefinitely sustainable. Dynamic rotation could improve cooling, but is quantified only by a calibrated Rev0 lumped model, not by corresponding verified Rev1 rotating-flow solutions.

10. Engineering discussion: what these results mean

10.1 Why adding stages changes the torque–loss balance

At fixed current, ten boards dissipate about twice the resistive loss of five boards at the same copper temperature. At fixed target torque, however, the current requirement decreases approximately inversely with K; resistive heating therefore scales approximately with R/K^2. This provides the physical explanation for the thermal record's observation that Rev1-D, though hotter at 1 A than Rev0, can be much cooler at a common 20 mN m torque request. [EM record p. 43; thermal record pp. 8–9]

For fixed T: I_peak = T/(1.5 K) ; P_Cu = (2/3) × R_phase × (T/K)^2

10.2 What Rev1-D wins and what it costs

Rev1-D offers the largest modelled fundamental K, highest motor constant and most favourable thermal torque comparison. It also has 48 rather than 16 magnets, two five-board stators, added rotor mass, an estimated 47–48 N rotor attraction and substantially higher back-EMF. At a fixed 12 V supply it has a lower voltage-limited speed than Rev0. The preferred design therefore depends on whether torque density, temperature margin, speed, parts count, cost and mechanical rigidity are the primary requirements.

10.3 The cooling assumption is consequential

Icepak uses still rotor(s), a sealed assumed housing, 30 °C local hood air, 1 m/s fan inlet and no radiation. A different mounting arrangement, venting, contact resistance, winding coverage or airflow could materially change absolute winding temperatures. The report itself warns that its turning-rotor convection correlation is uncalibrated. Thus the most defensible thermal statement is a comparative modelled limit under specified boundary conditions, not a final hardware rating. [Thermal record pp. 4, 11–12]

10.4 Magnet thermal margin is comparatively narrow

At the modelled 100 °C hottest-board limit, the record places N35 magnets at approximately 73–74 °C, below but close to an approximate 80 °C rating. This leaves little margin for elevated ambient temperature, uneven local heating and build differences. The record recommends considering higher-temperature magnet grades if testing reveals insufficient margin. [Thermal record pp. 1, 9]

11. Verification quality and remaining limitations

Workstream	Evidence of checks	Residual limitations
EM electrical	Conduction gates, phase-resistance balance, two-board series test, terminal and junction probes	Prototype fabrication/contact resistance not measured
EM magnetic	Reciprocity, harmonic/periodicity checks, superposition; Rev0 field-only gate +1.40% and 0.04°	Rev1-S2 copper-reference discrepancy 3–11%; no copper eddy/armature reaction
Thermal steady	R(T) iterations, energy/temperature scaling, physical trends, corrected mesh omissions	One reported final loss-change as high as 1.1%; assumed housing and air conditions
Thermal transient	Rev0 lumped model matches still-rotor Icepak rise within ±7%	Rotation-induced cooling not measured; no dedicated Rev1 rotating model

Neither report documents a completed prototype measurement of shaft torque, voltage, rotation-induced cooling, phase resistance versus temperature or actual steady-state efficiency. Any thesis conclusion should distinguish solved numerical observations, derived algebra, parameter estimates and assumptions. [EM record pp. 48–50; thermal record pp. 11–13]

12. Bench validation plan and practical recommendations

Electrical: measure phase resistance at a recorded winding temperature and compare with the predicted 1.389 / 2.777 ohm phase values plus leads and joints. Spin Rev0 at known speed and measure open-circuit fundamental phase rms back-EMF, expected approximately 1.444 V at 1000 rpm from the final EM model. Measure phase order, winding polarity and torque versus peak current when the mechanical rig and drive are safe. [EM record pp. 48–49]

Thermal: place thermocouples near the winding radius on boards near and far from rotor surfaces, instrument local air temperature and capture resistance versus time as an indirect winding-temperature estimate. Begin with current at or below the report’s stationary-rotor conservative limits, with appropriate temperature/current interlocks. Compare a locked-rotor heat-up case to a run under controlled rotation to calibrate air stirring. Do not deliberately operate Rev1 at 2 A

peak continuously under the stated enclosure conditions. [Thermal record pp. 12–13]

Mechanical: verify the 1 mm air gaps under assembled tolerances and magnetic load; assess bearing and shaft axial load paths; retain a protective enclosure and emergency stop. A numerical attraction estimate of nearly 48 N requires physical design attention before rotor operation. [EM record p. 40]

Modelling: repeat thermal solves once the actual housing is known, reconcile EM K revision in torque axes and operating-point setpoints, investigate the Rev1 S2 field-only/detailed discrepancy, and if resources permit cross-check one complete machine on high-memory hardware. A thermal transient or supported rotating-flow model would improve confidence in continuous ratings. [EM record pp. 48–50; thermal record pp. 11–13]

13. How to write the final report from these records

Thesis chapter	Core material
Methodology	Original PCB import/trace connectivity; AEDT gates; copper-FEM reciprocity; field-only FEM and region validation; MATLAB lumped dynamics; Icepak geometry/materials, boundary conditions and resistance-temperature iteration
Results	Final EM comparison and board contributions; magnetic steel/force metrics; MATLAB predicted speed and Cu-loss-limited efficiency; Icepak hotspot/magnet temperatures and original stationary-rotor limits
Discussion	Why cascades improve K and K_m; diminishing PCB stack contribution; thermal torque advantage; back-EMF speed cost; axial loads; region-bias diagnosis; housing/flow assumptions and Rev1 uncertainty
Conclusion	Evidence supports stronger EM coupling and improved modelled temperature-limited torque for cascaded designs under selected conditions; physical continuous ratings and full efficiency remain unverified
Recommendations	Physical testing, instrumentation, higher-temperature magnets if needed, mounting/air-gap verification, refined cooling model and further EM cross-validation

14. Suggested conclusion, appropriately qualified

Within the investigated geometry and operating assumptions, cascaded PCB axial-flux motor configurations provide substantially higher electromagnetic torque constants than the Rev0 single-stage baseline. The double-sided-centre-rotor Rev1-D configuration achieves the highest modelled torque constant, approximately 3.09 times Rev0, and the largest motor constant. The thermal simulations suggest that this advantage can translate into superior temperature-limited torque output even though the Rev1 designs carry more copper and tolerate lower sustained currents at fixed housing and cooling conditions. However, the quoted steady-state temperatures depend on an assumed enclosure and stationary-rotor boundary conditions, while Rev1 absolute electromagnetic coupling retains a remaining consistency discrepancy. Accordingly, these results provide a quantitative, simulation-based feasibility assessment and design ranking, rather than experimentally established continuous motor ratings.

15. Source register and provenance

Primary source A: PCB Axial-Flux Motor: Modelling Timeline and Methodology Record, 50 pages, dated 8 October 2026. Main sections used: project setup pp. 1–4; chronological development pp. 6–17; run register pp. 18–21; reduced method pp. 29–30; computational limit pp. 31–33; final field-only method pp. 34–37; Rev1 analysis pp. 37–40; MATLAB pp. 41–44; recommended thesis mapping and limitations pp. 45–50.

Primary source B: PCB Axial-Flux Motor: Thermal Analysis Report, 14 pages, dated 8 October 2026. Main sections used: summary pp. 1–2; Icepak geometry/materials/conditions pp. 3–4; Rev0 results pp. 5–6; Rev1 results pp. 7–8; continuous-torque comparison p. 9; verification pp. 10; assumptions pp. 11–12; recommendations and scripts pp. 12–14.

Derived in this companion only: current-limit torque conversions 51.2, 83.1 and 118.5 mN m using $T = 1.5\ K_{final}\ I_{peak}$. These are labelled conditional algebraic calculations and were not present as final Icepak-derived continuous ratings in either source. Other values are identified as source report numbers or qualitative interpretation.